UNIT 2 · 8 hours

Type of Systems and Static Error Constants



Time Domain Analysis of Control Systems

Control Systems (22EEG53)
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Learning objectives

- Define the type of a system
- Define position, velocity and acceleration error constants
- Compute steady-state error for step, ramp and parabolic inputs
- Relate type to error

Syllabus context

Time domain analysis of control systems: standard test signals, time response of zero, first and second order systems, time domain specifications, type of systems, static error constants and steady-state error, generalized error series.

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Type of a system

G(s)H(s) = K Π(1 + Tz s) / [s^N Π(1 + Tp s)]

open-loop transfer function

Type = N

number of poles at s = 0

Error: e(t) = r(t) − c(t)

unity feedback

- Type 0: no pole at the origin; type 1: one (an integrator); type 2: two.
- Type, not order, determines steady-state accuracy.

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Static error constants

Constant	Definition	Input
Position Kp	$\lim_{s\to 0} G(s)H(s)$	Step
Velocity Kv	$\lim_{s\to 0} s\,G(s)H(s)$	Ramp
Acceleration Ka	$\lim_{s\to 0} s^2\,G(s)H(s)$	Parabola

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Steady-state error: unity feedback

Type	Step (1)	Ramp (t)	Parabola ($t^2/2$)
0	1/(1 + Kp)	∞	∞
1	0	1/Kv	∞
2	0	0	1/Ka
3	0	0	0

ess = lim s→0 s R(s)/(1 + G(s)): valid only for a stable closed-loop system.

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Numerical 1: type 1 system

Problem Unity feedback with G(s) = 10/[s(s + 2)]. Find type, Kp, Kv, Ka and ess for unit step, unit ramp.

- Type 1 (one pole at origin)
- Kp = ∞; Kv = lim s·G = 10/2 = 5; Ka = 0
- ess (step) = 0; ess (ramp) = 1/Kv = 0.2

Answer: Type 1; Kv = 5; ess = 0 (step), 0.2 (unit ramp)

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Numerical 2: type 0 and type 2

Problem (a) $G = 20/[(s+1)(s+4)]$; (b) $G = 10(s+1)/[s^2(s+4)]$, unity feedback. Find the error for a unit step (a) and a unit parabola (b).

- (a) Type 0: $K_p = 20/4 = 5 \Rightarrow e_{ss} = 1/(1 + 5) = 0.167$
- (b) Type 2: $K_a = \lim s^2G = 10(1)/4 = 2.5$
- e_{ss} (parabola) $= 1/K_a = 0.4$; e_{ss} (ramp) $= 0$

Answer: (a) 0.167; (b) 0.4

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Remarks

- Unity feedback assumed; for H ≠ 1 use e = r − b or the actuating error carefully.
- K enters the constants: a larger gain gives a smaller error.
- Stability limits how large K can be: accuracy versus stability trade-off.
- Compensators (Unit 5) improve accuracy without destroying stability.

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Key takeaways

- Type = number of poles at the origin of G(s)H(s)
- Kp = lim G; Kv = lim sG; Ka = lim s^2G
- ess = 1/(1 + Kp), 1/Kv, 1/Ka for step, ramp, parabola
- Higher type ⇒ lower error but harder stability

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Practice questions

Q1. Define Kp, Kv and Ka.

Q2. G = 50/[s(s+5)(s+10)]: find Kv and ramp error.

Q3. A type 0 system has Kp = 9: find the step error.

Q4. Tabulate steady-state error versus type.